6. Show that $\dfrac{(n+1)(n+2)(n+3)\ldots(n+r)}{\lfloor r}$ is a whole no.

7. If ${}^{28}C_{2r} : {}^{24}C_{2r-4} = 225 : 11$, find ${}^{r}C_5$.

8. If ${}^{n}C_{12} = {}^{n}C_8$, find ${}^{n}C_{17}$.

9. If ${}^{n}C_x = 56$ and ${}^{n}P_x = 336$, find n and x.

10. Determine n if $C(2n, 3) : C(n, 3) = 11 : 1$.

11. Find the value of ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$.

12. Evaluate $C(19, 17) + C(19, 18)$.

13. Define combination ? What is the difference between a permutation and combination.

14. Prove that ${}^{n}C_r \times {}^{r}C_s = {}^{n}C_s \times {}^{n-s}C_{r-s}$. **(P.U. B.C.A.–I 2001)**

15. If ${}^{n}P_r = {}^{n}P_{r+1}$ and ${}^{n}C_r = {}^{n}C_{r-1}$ find values of n and r.

16. Evaluate

(*i*) $C(5, 2) + C(5, 3)$

(*ii*) $C(3, 0) + C(4, 0) + C(5, 0)$

(*iii*) $C(1, 1) + C(2, 2) + C(3, 3) + C(4, 4)$

(*iv*) $C(5,5) + C(6, 5) + C(7, 5)$

ANSWERS

1. (*i*) 45, (*ii*) 50 **2.** (*i*) 462, (*ii*) 34220 **4.** $n = 14, r = 4$ **7.** 21

8. 1140 **9.** $x = 3, n = 8$ **10.** 6 **11.** 270725

12. 190 **16.** (*i*) 20, (*ii*) 3, (*iii*) 4, (*iv*) 14

PRACTICAL EXAMPLES ON COMBINATIONS

Example 1. From a class of 32 students, 4 are to be chosen for a competition. In how many ways can this be done ?

Sol. Total no. of students = 32

We are to select 4 students

$\therefore$ Required no. of ways = ${}^{32}C_4$

$$= \frac{\lfloor 32}{\lfloor 4 \lfloor 28}$$

$$= \frac{32 \times 31 \times 30 \times 29 \lfloor 28}{4.3.2.1. \lfloor 28} = 3560.$$

Example 2. In how many can 4 sports persons be selected from a group of 10.

Total no. of persons = 10

Persons to be selected = 4

Required no. of ways = ${}^{10}C_4$

$$= \frac{\lfloor 10}{\lfloor 4 \lfloor 6}$$

$$= \frac{10.98.7 \lfloor 6}{4.3.2.1. \lfloor 6}$$

$$= 210$$

Example 3. In how many ways can a committee be selected from 15 persons if the committee have : (*i*) 3 members, (*ii*) 13 members.

Sol. (*i*) Required no. of committees = ${}^{15}C_3$

$$= \frac{\lfloor 15}{\lfloor 3 \lfloor 12}$$

$$= \frac{15.14.13. \lfloor 12}{3.2.1. \lfloor 12} = 455$$

(*ii*) No. of committes = ${}^{15}C_{13} = \frac{\lfloor 15}{\lfloor 13 \lfloor 2}$

$$= \frac{1514. \lfloor 3}{\lfloor 13\, 2.1}$$

$$= 105$$

Example 4. How many lines can be drawn through 6 points on a circle ?

Sol. A line is drawn if we join any 2 points. We have six points.

$\therefore$ Required no. of ways = ${}^{6}C_2 = 15$

Example 5. A polygon has 44 diagonals, find the numbers of its sides.

Sol. Let the polygon have n sides

$\therefore$ No. of diagonals = ${}^{n}C_2 - n$

Given ${}^{n}C_2 - n = 44$

$$\frac{n(n-1)}{2} - n = 44$$

$\Rightarrow$ $m^2 - n - 2n = 88$

$\Rightarrow$ $n^2 - 3n - 88 = 0$

$\Rightarrow \quad (n-11)(n+8)=0$

$\Rightarrow \quad n = 11$ or $n = -8$

$\therefore$ No.of sides $= 11$

Example 6. How many words containing 4 consontants and 3 vowels can be formed from constants and 4 vowels.

Sol. No. of ways in which 4 consonants can be selected out of $6 = {}^6C_4$

No. of ways in which 3 vowels can be selected out of $4 = {}^4C_3$

Now seven letters can be arranged in ${}^7P_7$ *i.e.* $\lfloor 7$ ways.

$\therefore$ Total no. of ways $= {}^6C_4 \times {}^4C_3 \times \lfloor 7$

$$= \frac{6.5}{2.1} \times \frac{4}{1} \times 5040$$

$$= 302400$$

Example 7. Out of 6 boys and 4 girls a committee of 6 is to be formed. In how many ways can this be done if the committee contains :

(*i*) exactly 2 girls ? (*ii*) at least 2 girls ?

Sol. (*i*) Each committee contains 4 boys and 2 girls. Number of ways in which 4 boys can be chosen out of $6 = {}^6C_4$

No. of ways in which 2 girls can be chosen out of $4 = {}^4C_2$.

$\therefore$ Total no. of ways $= {}^6C_4 \times {}^4C_2 \quad = \frac{6.5}{2.1} \times \frac{4.3}{2.1}$

$$= 90$$

(*ii*) Because each committee consists at least 2 girls.

$\therefore$ Committee can have 2 or 3 or girls.

(*i*) If two girls and 4 boys are to be selected. Number of ways in which 4 boys can be chosen out of $6 = {}^6C_4$

No. of ways in which 2 girls can be chosen out of $4 = {}^4C_2$

$\therefore$ Total no. of ways $= {}^6C_4 \times {}^4C_2 \quad = 90$

(*ii*) **If 3 girls and 3 boys are to be selected.** No. of ways in which 3 boys can be chosen out of 6 and 3 girls out of 4 can be chosen $= {}^6C_3 \times {}^4C_3 = 80$

(*iii*) **If 4 girls and 2 boys are to be selected.** No. of committee having 2 boys and 4 girls $= {}^6C_2 \times {}^4C_4$ $= 15$.

$\therefore$ Total no. of committees $= 90 + 80 + 15$

$$= 185$$

Example 8. The judiciary committee at a college is made up of 3 faculty members and 4 students. If 10 faculty members and 25 students have been nominated for committee, how many committee are possible ?(*P.T.U. B.C.A.–I 2005*)

Sol. No. of ways in which 3 faculty members are selected out of 10 = $^{10}C_3$.

and ways in which 4 students can be selected 25 students = $^{25}C_4$

$$\text{Total no. of ways} = {}^{10}C_3 \times {}^{25}C_4 = \frac{\lfloor 10}{\lfloor 7 \lfloor 3} \times \frac{\lfloor 25}{\lfloor 4 \lfloor 21}$$

$$= \frac{10.9.8.\lfloor 7}{3.2.1.\lfloor 7} \times \frac{25.24.23.22.\lfloor 21}{\lfloor 21\, 4.3.2.1}$$

$$= \frac{10.9.8}{3.2.1} \times \frac{25.24.23.22}{4.3.2.1}$$

$$= 1518000$$

Example 9. How many 6 digits numbers without repetition of digit are such that the digits are non zero and 1 and 2 donot appear consecutive in either order ? (Pbi.U.B.C.A. 2006)

Sol. Since we are to take non zero digits.

∴ We have total 9 digits (1, 2, 3, ..., 9)

6 digits out of 9 can be selected in $^9C_6$ ways Further 6 digits can be arranged in $\lfloor 6$ ways

$$\therefore \quad \text{Total number of ways} = {}^9C_6 \times \lfloor 6$$

$$= \frac{\lfloor 9}{\lfloor 3 \lfloor 6} \times \lfloor 6$$

$$= \frac{\lfloor 9}{\lfloor 3}$$

$$= \frac{9.8.7.6.5.4.\lfloor 3}{\lfloor 3}$$

$$= 60480$$

From this, we exclude those no. in which 1, 2 are taken consecutively.

Since 1, 2 are taken so remaining 4 digits can be selected from remaining 7 digits in $^7C_4$ ways and taking (1, 2) as single digit.

Now these no. can be arranged in $\lfloor 5 \times \lfloor 2$ ways.

Total no. of ways in which 1, 2 occur together

$$= \frac{\lfloor 7}{\lfloor 3 \lfloor 4} \times \lfloor 5 \times \lfloor 2 = 8400.$$

$\therefore$ Required no. of ways = 60480 – 8400

= 52080

Example 10. In how many ways can a teacher choose one or more students from six eligible students ? **(P.U. B.C.A.–I 2002)**

Sol. Teacher can choose one or two or three or four or five or six students

No. of ways in which one student can be selected out of 6 = ${}^6C_1$

No. of ways in which two student can be selected out of 6 = ${}^6C_2$

No. of ways in which three student can be selected out of 6 = ${}^6C_3$

No. of ways in which four student can be selected out of 6 = ${}^6C_4$

No. of ways in which five student can be selected out of 6 = ${}^6C_5$

No. of ways in which six student can be selected out of 6 = ${}^6C_6$

$$\text{Total no. of ways} = {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$$

$$= 2^6 - 1$$

$$= 63$$

Example 11. How many subsets of {1, 2, ..., 10} contain at least 7 elements ?

(P.T.U. B.Tech. Dec. 2005)

Sol. Total no. of elements =10

$\therefore$ Subset can contain 7 or 8 or 9 or 10 elements

$\therefore$ No. of ways in which subsets are made with

7 elements out of 10 = ${}^{10}C_7$

No. of ways in which subsets are made with element out of 10 = ${}^{10}C_8$.

No. of ways in which subsets are made with 9 elements out of 10 = ${}^{10}C_9$

No. of ways in which subsets are made with 10 elements out of 10 = ${}^{10}C_{10}$

$\therefore$

$$\text{Total no. of ways} = {}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}$$

$$= \frac{\lfloor 10}{\lfloor 7 \lfloor 3} + \frac{\lfloor 10}{\lfloor 8 \lfloor 2} + \frac{\lfloor 10}{\lfloor 9 \lfloor 1} + \frac{\lfloor 10}{\lfloor 10 \lfloor 0}$$

$$= \frac{10.9.8.\lfloor 7}{\lfloor 7 .3.2.1} + \frac{\lfloor 10\, 9 \lfloor 8}{\lfloor 8 .2.1} + \frac{10.\lfloor 9}{\lfloor 9} + \frac{\lfloor 10}{\lfloor 10}$$

$$= 120 + 45 + 10 + 1$$

$$= 176$$

Example 12. A bag contains 5 red marbles and 6 white marbles. Find out in how many ways

(*i*) 4 marbles can be drawn ?

(*ii*) If 1 marble is red and 2 are white ?

480 – 8400

0

hoose one or more students from six eligible (P.U. B.C.A.–I 2001)

ive or six students

$5 = {}^6C_1$

$= {}^6C_2$

$= {}^6C_3$

C_4

$+ {}^6C_5 + {}^6C_6$

ments ?

T.U. B.Tech. Dec. 2005

of 4 marbles must be of some colour.

No. of red marbles = 5

No. of white marbles = 6

Total no. of marbles = 11

(i) No. of ways in which 4 marbles can be drawn out of 11 = ${}^{11}C_4$

$$= \frac{\lfloor 11}{\lfloor 4 \lfloor 7}$$

$$= \frac{11.10.9.8.\lfloor 7}{4.3.2.1.\lfloor 7}$$

$$= 330$$

(ii) Required no. of ways $= {}^5C_2 \times {}^6C_2$

$$= \frac{5.4}{2.1} \times \frac{6.5}{2.1}$$

$$= 10 \times 15$$

$$= 150$$

(iii) If 4 marbles are of some colour these are either red or white

$\therefore$ Required no. of ways $= {}^5C_4 + {}^6C_4$

$$= 5 + 6$$

$$= 11$$

Example 13 Suppose that there are n-people in a room $n > 1$ and that they all shake hands with one another. prove that $n(n-1)/2$ handshakes will occur. **(P.T.U. B.Tech. Dec. 2005)**

Sol. When two persons shake hands, it is taken as one hand shake.

$\therefore$ Total no. of hands slakes $= {}^nC_2$

$$= \frac{\lfloor n}{\lfloor 2 \lfloor n-2}$$

$$= \frac{n(n-1)\lfloor n-2}{2.1\lfloor n-2}$$

$$= \frac{n(n-1)}{2}$$

Example 14. In how many ways can 12 things be equally divided (i) between 2 persons, (ii) into 2 heaps ?

Sol. (i) Since 12 things are equally divided between 2 persons

$\therefore$ groups are distinct

$$\therefore \quad \text{Required no. of ways} = \frac{\lfloor 12}{\lfloor 6 \lfloor 6} = \frac{12.11.10.9.8.7.\lfloor 6}{6.5.4.3.2.1.\lfloor 6}$$

$$= 924$$

(*ii*) Since 12 things are to be equally divided into two heaps.

$\therefore$ no. distinct an can *i.e.* made between two groups

$$\therefore \quad \text{Required no. of ways} = \frac{1}{\lfloor 2}\frac{\lfloor 12}{\lfloor 6 \lfloor 6} = \frac{1}{2}(924) = 462$$

Example 15. A man has 7 friends. In how many ways can be invite one or more of them to a party ?

Sol. The man can invite 1 or 2 or 3 or ... or 7 friends

$$\therefore \quad \text{Required no. of ways} = {}^7C_1 + {}^7C_2 + {}^7C_3 + \ldots + {}^7C_7$$

$$= 2^7 - 1$$

$$= 128 - 1$$

$$= 127$$

Example 16. Determine the number of 5-card combination out of 52 cards if atleast one of the 5 cards has to be a king.

Sol. The possibilities are

(*i*) 1 king and 4 non kings

(*ii*) 2 kings and 3 non kings

(*iii*) 3 kings and 2 non kings

(*iv*) 4 kings and 1 non king

$$\therefore \quad \text{Required no. of ways} = {}^4C_1 \times {}^{48}C_4 + {}^4C_2 \times {}^{48}C_3 + {}^4C_3 \times {}^{48}C_2 + {}^4C_4 \times {}^{48}C_1$$

$$= 886656$$

Example 17. The English alphabet has 5 vowels and 21 constants. How many words with 2 different vowels and 2 different consontants can be formed from the alphabets ?

Sol. No. of ways in which 2 vowels can be selected out of 5 = ${}^5C_3$

No. of ways in which 2 consontants can be selected out of 21 = ${}^{21}C_2$

lowers can be arranged in $\lfloor 4$ ways.

$$\text{Total no. of ways} = {}^5C_3 \times {}^{21}C_3 \times \lfloor 4$$

$$= 10 \times 210 \times \lfloor 4$$

$$= 2100 \times 24$$

$$= 50400$$

ple 18. Among $(3n + 1)$ objects, n of them are identical. Find the no. of ways to select n of these $(3n + 1)$ objects.

(G.N.D.U. M.Sc.-I 2009)

$$\text{Total no. of objects} = 3n + 1$$

of ways in which n objects can be selected out of $3n + 1 = {}^{3n+1}C_n$

of them are identical

$$\text{Required no. of ways} = \frac{{}^{3n+1}C_n}{\lfloor n}$$

$$= \frac{\dfrac{\lfloor 3n+1}{\lfloor n \lfloor 3n+1-n}}{\lfloor n}$$

$$= \frac{\lfloor 3n+1}{(\lfloor n)^2 \lfloor 2n+1}$$

Example 19. In an examination, Sunita has to select 4 questions from each part. There are 6, 7 questions in Part–I, Part–II and Part–III respectively. What is the number of possible ations in which she can choose the question ?

Sol. No. of questions in Part-I = 6

No. of questions in Part-II = 7

No. of questions in Part–III = 8

No. of ways in which is questions selected out of 6 = ${}^6C_4$

No. of ways in which is questions selected out of 7 = ${}^7C_4$

No. of ways in which is questions selected out of 8 = ${}^8C_4$

$$\therefore \quad \text{Required no. of ways} = {}^6C_4 \times {}^7C_4 \times {}^8C_4$$

$$= 36750$$